

Research Article

Addressing Class Imbalance Problem in Health Data Classification: Practical Application From an Oversampling Viewpoint

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While analyzing health data is important for improving health outcomes, class imbalance in datasets poses major challenges to machine learning classification models. This work, therefore, considers the class imbalance problem in stroke prediction using models such as K-nearest neighbors, support vector machine, logistic regression, random forest, and decision tree. This work balances the stroke dataset, thereby enhancing model performance, through various oversampling strategies: random oversampling (RO), ADASYN, SMOTE, and SMOTE-Tomek. Compared to the results of the imbalanced dataset, all applied oversampling techniques enhanced the correct classification of stroke events by the ML model. Among these, RO-SVM with RBF kernel was the best in terms of sensitivity, specificity, G-mean, F1-score, and accuracy values, offering the highest results with respective values of 89.87%, 94.91%, 92.36%, 89.64%, and 89.87%. After applying oversampling techniques, all the machine learning classifications were good enough to classify stroke status, especially for the minority class. This study has highlighted the importance of class imbalance issues in health datasets. Precise detection of instances of minority classes can be enhanced considerably by employing classification models with the implementation of hybrid strategies to effectively solve class imbalance issues, which, in turn, will help improve healthcare outcomes. Further research in integrating more advanced deep learning techniques into other health datasets with imbalances is encouraged to further validate or refine class imbalance approaches, as effective handling of imbalanced classes can substantially promote predictive model performance in the analysis of healthcare.

Keywords: class imbalance; health data classification; oversampling techniques; stroke prediction

1. Introduction

Analysis of health data plays a major role in understanding and improving healthcare. Insights from analyzing health-related datasets may facilitate the implementation of public health policy, enhance clinical decision-making, and promote better patient care [1]. However, one of the most common challenges while working on health-related datasets is the issue of class imbalance [2]. Most machine

learning (ML) classification models assume that there is a uniform distribution within the target variable classes. This assumption is nearly always violated in disease diagnostics since the number of representatives from one class (e.g., healthy patients) significantly outnumbers the instances of the other class (e.g., patients with a specific disease). Class imbalance becomes apparent if the distribution of distinct classes is not uniform; for instance, disease cases outnumber nondisease cases [3]. This is an imbalance that may affect the

learning process of any ML algorithm from the data. For instance, if in a dataset, 95% of the instances are labeled as nondisease and only 5% as disease, then a model may be biased in predicting the majority class. The consequence is that performance for a model might be fine on an overall accuracy level but might miss accurately identifying the minority class that often happens to be from the class of highest interest concerning medical diagnostics. Class imbalance is statistically considered a major problem because often the models are making poor predictions [4]. In such cases, when the model is exposed to an imbalanced dataset, it learns to predict more about the majority class, not because of anything special but rather simply because of being exposed more often. The result is usually a very high accuracy score, which is misleading as it might not detect instances of the minority class. For instance, in a disease diagnosis scenario, missing out on detecting the disease (minority class) can have serious implications, including delayed treatment and poor patient outcomes.

Class imbalance is an important issue not only for model performance improvement but also for making the model fair and reliable. Techniques to handle class imbalance include data resampling methods, algorithmic approaches, and performance metrics specifically proposed for imbalanced data. However, resampling methods such as oversampling and undersampling are usually adopted among them. Oversampling generates additional instances of the minority class [5] and undersampling involves reducing the number of instances in the majority class [6]. Both techniques aim at a balanced distribution of classes for easy learning by the algorithm. Various oversampling techniques involve the generation of artificial examples of the minority class as an interpolation between the different examples of the minority class, such as the synthetic minority oversampling technique or SMOTE. However, methods for undersampling reduce the number of a few kinds of the majority class in order to balance the dataset: Random undersampling and one of the most important editions of near miss focus mainly on how to select the representative subset from the majority class that maintains the meaningful characteristics of the dataset. While effective, the process of undersampling may lead to the loss of valuable information from the majority class and should hence be applied with caution [7]. It is hence important to select an appropriate resampling technique concerning the specific characteristics of the dataset and the problem at hand. For example, when the minority class is extremely underrepresented, oversampling techniques such as SMOTE may be better suited [8]. On the other hand, in case the majority class contains a lot of redundant information, undersampling techniques may be more suitable for use [9].

Class imbalance may cause considerable problems in machine learning, leading to the generation of biased models. This is especially problematic in applications in healthcare, where the minority group often represents significant conditions, such as the presence of a disease. A model that fails to properly predict these events of a minority class can lead to missed diagnosis and delayed or inappropriate treatments. Consequently, reducing model bias,

one way being through resampling, or the modification of class weights, is highly important in capturing overall performance. Overfitting is another problem brought about by class imbalance, albeit this time on the side of the majority class. Normally, overfitting can be understood to mean an instance when a model happens to have learned the noisy and outlying data excessively well during training, while it may have learned real patterns. An imbalanced dataset results in models which will tend to prefer the memorization of majority class examples since those dominantly constitute the training phase [10]. An overwhelming concentration with respect to the majority class results in a model performance which will do quite exceptionally during the training process, though it fails when generalizing for data not previously seen. Due to this, the minority class, which is already a subpopulation, gets even less attention, and hence the poor performance of the model in predicting such cases. Overfitting diminishes predictive capability while reducing reliability and strength for deployment into real-life applications. In healthcare, where highly accurate predictions are expected of models, a model overfit to the majority class could diagnose incorrectly or variably and put patients' lives in jeopardy. To handle overfitting, some techniques include the methods of cross-validation and regularization, along with working with balanced datasets using the method of resampling [11].

Class imbalance also causes reduced sensitivity, or recall, of the classification model with regard to the minority class. In imbalanced datasets, the performance of the model usually highly compromises the correct labeling of instances from the minority class, since the model is biased toward predicting the majority class. This reduced sensitivity means that many instances of the minority class, for example, positive disease cases, may go undetected. This becomes particularly dangerous in healthcare, as it could lead to missed diagnoses, the underestimation of prevalence, and the lack of proper medical interventions. This could be improved using resampling techniques in balancing the dataset, adjusting the decision threshold, or utilizing other metrics for evaluation which take class imbalance into consideration, such as the F1-score or the area under the precision–recall curve [12]. The model is improved in sensitivity because it acquires reliability and efficacy by better detection of critical instances from minority classes since most of the members belong to the positive class that always gives better general results in applications, such as healthcare one's [13]. In anomaly detection [14–17], oversampling techniques can be a vital statistical tool.

It is of interest to know that once the various resampling techniques are applied, evaluating model performance carefully is crucial. Traditional metrics such as accuracy can be misleading when dealing with imbalanced datasets. Instead, metrics such as precision, recall, F1-score, the area under the precision–recall curve, and the area under the ROC curve (AUC-ROC) provide a better understanding of how well the model performs with respect to both classes [18]. Cross-validation techniques can also help ensure that the model's performance is robust and not dependent on a particular train–test split. By addressing class imbalance

effectively, the performance of predictive models in healthcare analysis can be significantly improved. Better model performance translates to more accurate predictions, which can lead to improved patient outcomes. For example, a model that accurately identifies patients at risk of developing a particular disease can enable timely interventions and better management of the disease. Similarly, models used for diagnostic purposes can help in early detection, leading to better treatment options and improved survival rates. This study aimed to provide a general overview of the stroke status data and compare the performance of K-nearest neighbor (KNN), support vector machine (SVM), logistic regression, random forest (RF), and decision trees (DTs) classification on datasets without oversampling against datasets with various oversampling techniques including RO, ADASYN, SMOTE, and SMOTE combined with Tomek links (SMOTE-Tomek).

In this study, several contributions to the field of health data classification in the context of stroke prediction are made. First, the performance of various oversampling techniques, random oversampling (RO), adaptive synthetic sampling (ADASYN), SMOTE, and SMOTE-Tomek, on the performance of ML models are assessed. Second, a detailed comparison of several ML algorithms, including KNNs, SVM, logistic regression, RF, and DTs, is provided to identify which combinations of models and oversampling techniques yield the best outcomes for predicting strokes. This comprehensive evaluation offers valuable insights for improving healthcare diagnostics and interventions.

The remainder of the paper is organized as follows. Section 2 presents the literature review and related works. Section 3 discusses the data and methods used for the study. Section 4 provides the mathematical framework of the study. Section 5 discusses the results of the study, whilst Section 6 concludes the study and provides recommendations for further work.

2. Literature Review/Related Works

To tackle the issue of class imbalance [19], we categorize the existing solutions into three main approaches: data-level methods, algorithm-level methods, and hybrid methods. Data-level methods, such as oversampling and under-sampling, focus on adjusting the class distribution within the dataset itself. Hybrid/ensemble methods, which include techniques such as bagging and boosting, combine multiple algorithms to improve classification performance on imbalanced datasets. By aggregating the outputs of several classifiers, ensemble methods can reduce variance and bias, leading to more robust predictions. Hybrid methods, which integrate both data-level adjustments and algorithmic modifications, offer a comprehensive solution to the class imbalance problem with higher accuracy. For example, SMOTEBoost combines the SMOTE technique with boosting, generating synthetic examples of the minority class while also adjusting the model's learning process to account for the class imbalance. By reviewing 24 studies published between 2003 and 2019, the authors in [20] provided a comprehensive review of

recent methodologies developed to tackle the issue of class imbalance in large datasets. The study focuses on high-class imbalance, where the ratio of majority to minority classes can be as extreme as 10,000:1, which poses significant challenges for ML models. Existing approaches were categorized into data-level methods (such as oversampling and under-sampling) and algorithm-level methods (such as cost-sensitive learning and hybrid/ensemble techniques). Through the survey of studies conducted over the last eight years, the strengths and limitations of these methods were highlighted, particularly in the context of big data environments such as Apache Hadoop and Apache Spark. The review concludes that while some methods such as random oversampling (RO) and cost-sensitive algorithms show promise, there are inconsistencies and gaps in the current research. The study recommended more comprehensive, comparative studies to better understand the effectiveness of these approaches in different big data scenarios.

2.1. Advanced Techniques and Frameworks for Addressing Class Imbalance. In response to the challenges posed by class imbalance, more sophisticated frameworks and methodologies have been developed to improve the handling of imbalanced medical datasets. The use of advanced deep learning architectures and ensemble methods offers promising solutions. For instance, the study involving deep ensemble learning for pneumoconiosis detection employs multiple deep learning models in an ensemble setting to average out biases and improve the robustness of predictions [21]. This method not only addresses the imbalance but also reduces the variance among model predictions, resulting in a more reliable diagnostic tool. Furthermore, the innovative application of feature engineering in ovarian cancer prediction using the CA125 biomarker demonstrates how ML can be finely tuned to enhance model performance in imbalanced settings [22]. By selecting and engineering specific features that are highly indicative of minority classes, models can achieve greater sensitivity to these classes without the need for extensive data collection, which is often impractical in clinical environments.

Likewise, in addressing the challenge of class imbalance in medical imaging, which often leads to biased predictions favoring overrepresented classes, the authors in [23] tackle this issue by employing data augmentation techniques such as image rotation, scaling, and flipping to balance the distribution of skin lesion classes in the HAM10000 dataset. It evaluates the performance of three advanced deep learning models: AlexNet, InceptionV3, and RegNetY-320, with RegNetY-320 showing the best performance in terms of accuracy, F1-score, and ROC curve metrics. The research highlights the importance of hyperparameter tuning, especially the learning rate, and demonstrates that lower rates can significantly enhance model training and predictive accuracy. The findings from [23] not only improve the diagnostic capabilities of skin cancer detection models but also provide a framework that can be applied to other medical imaging challenges involving imbalanced data, paving the way for further studies to refine these techniques and explore new approaches in model training and architecture design.

In the same manner, the authors in [24] delve into the prevalent issue of class imbalance in medical datasets, specifically using lung cancer data to evaluate the effectiveness of various resampling methods in improving the predictive performance of ML models. The focus is on three broad categories of resampling techniques: undersampling, oversampling, and hybrid methods, alongside traditional classifiers such as logistic regression, RF, and LinearSVC. The study finds that oversampling techniques generally provide a more stable and higher performance in terms of area under the curve (AUC) metrics, compared to undersampling and hybrid methods, which are sometimes prone to discarding useful information or overfitting. Specifically, RO and SMOTE are highlighted for their effectiveness in managing the class distribution more favorably without significant data loss or overfitting, making them particularly suitable for medical data where preserving all potential signals is crucial. This study significantly contributes to the field by systematically comparing the impact of various resampling strategies on model accuracy and stability, providing a valuable reference for researchers and practitioners working with imbalanced datasets in healthcare and other domains. It highlights the importance of choosing the appropriate resampling technique based on the characteristics of the data and the desired outcomes of the analysis.

2.2. Enhanced Deep and ML Innovations in Medical Diagnosis. Recent advancements in deep learning have significantly enhanced the accuracy and efficiency of medical imaging and disease diagnosis, particularly in cancer detection and neurological assessments. The integration of convolutional neural networks (CNNs) and transfer learning approaches has been pivotal. For instance, the development of specialized CNN architectures such as ConvADD for Alzheimer's disease showcases a tailored approach to neuroimaging, allowing for robust detection capabilities that surpass traditional imaging techniques [25]. Similarly, deep transfer learning models applied to MRI images for brain tumor classification demonstrate how pretrained networks can be effectively adapted to medical imaging, significantly improving diagnostic accuracy by leveraging learned features from extensive image datasets [26].

Parallel to image-based diagnostics, ML innovations have made significant strides in utilizing nonvisual data for disease detection, such as audio features for dementia recognition. Analyzing speech patterns and employing ML algorithms to detect early signs of dementia exemplifies how alternative data forms can provide critical diagnostic insights, particularly for conditions where early detection significantly alters treatment outcomes [27]. These methodologies highlight the shift towards comprehensive diagnostic systems that incorporate a variety of data inputs, reflecting a broader move towards multimodal disease detection frameworks.

2.3. Review of Machine and Deep Learning Techniques for Stroke Prediction. Table 1 presents a summary of the related literature relevant to the current study.

3. Data and Methods

The data used for the study are openly available on GitHub and can be assessed at the repository at <https://www.kaggle.com/datasets/mexwell/heart-disease-dataset/data>. The total number of stroke instances was 4891. Stroke is the target and is classified as “no stroke” (0) and “stroke” (1). The number of individuals for each “no stroke” and “stroke” status is 4733 and 248, respectively. The description of the stroke data is given in Table 2.

Before applying the various classifiers, the dataset was divided into two parts: a training subset consisting of 80% of the data ($n = 3984$) and a testing subset comprising the remaining 20% ($n = 997$). The training subset was used to construct the ML models: KNN models with varying k (3, 5, 7, 9, 11, 13, and 15), SVM with linear and radial basis function (RBF) kernels, logistic regression, RF, and DTs, whereas the testing subset was employed to evaluate the predictive accuracy of these models. Algorithms that involve distance calculations, such as KNN can be sensitive to the magnitude of the features [35]. Large differences in scales can lead to numerical instability and poor performance [36]. By standardizing the data, the risk of numerical instability is mitigated, and distance metrics are made meaningful. Before training the models, z-score normalization was adopted in this study to ensure that all features contribute equally to the final models, preventing dominance by features with larger scales and helping in faster convergence and numerical stability of the results.

In this study, optimal hyperparameters for the various ML models were chosen using grid search cross-validation. The optimal hyperparameters obtained by the algorithms are displayed as follows: For the RF and DT classifiers, $max_depth = 5$, $n_estimators = 100$, $min_samples_leaf = 2$, and $min_samples_split = 10$. For the logistic regression, $penalty = l2$, $max_iteration = 100$, $C = 2$, and $solver = liblinear$. For the SVM, $gamma = 1$ and $C = 2$. These parameters were selected to optimize the performance of each model. The grid search was utilized to systematically explore multiple combinations of parameters, determining which configuration provides the best results based on predefined metrics. The sensitivity of these parameters to the performance of the algorithms was significant, as evidenced by the detailed performance analysis included in the study. Key performance metrics such as sensitivity, specificity, G-mean, precision, F1-score, and accuracy were used to evaluate each model under different parameter settings. The optimal settings were those that balanced the detection of both minority and majority classes effectively. After successfully training and building the ML models, the predictive power of ML classifications on datasets without oversampling against datasets with various oversampling techniques, namely, RO, ADASYN, SMOTE, and SMOTE-Tomek links, which helped solve the class imbalance problem, was compared. The working architecture adopted for the study is depicted in Figure 1.

Figure 1(a) illustrates the concept of oversampling techniques, where additional copies of the minority class in a dataset are created to balance class distribution to help mitigate the issue of class imbalance in ML models.

TABLE 1: Summary of literature review on machine and deep learning techniques for stroke prediction.

Authors	Year	Techniques adopted	Key findings
[28]	2024	Logistic regression, decision trees, random forests, gradient boosting, SVM, and hybrid model (SMOTE + random forests)	Ensemble methods outperform individual classifiers; hybrid models deliver superior performance
[29]	2022	Logistic regression, decision trees, and neural networks	ML models can handle complex interactions and provide accurate predictions
[30]	2024	SMOTE-based oversampling coupled with logistic regression, decision trees, random forests, gradient boosting, AdaBoost, XGBoost, support vector machine (SVM), K-nearest neighbors (KNN), Naive Bayes (NB), bagging classifier (BC), and voting classifier (VC)	SMOTE-bagging techniques underline the possible role of advanced machine learning techniques in building feasible predictive tools for detecting stroke incidents in their incipient stage
[31]	2024	Decision tree, logistic regression, neural network (NN), random forest (RF), and NN-RF	SMOTE and ADASYN have proven to be viable tools to handle imbalanced problems for better evaluation metrics. Also, the hybrid neural network and random forest (NN-RF) utilizing the balanced dataset by ADASYN oversampling can enhance reliable stroke prediction
[32]	2022	Random downsampling coupled with decision tree, random forest, and neural network	A perceptron neural network, utilizing four significant risk factors (age, heart disease, average glucose level, and hypertension) demonstrated the highest accuracy and lowest miss rate in predicting stroke occurrences, compared to other models that used all available input features
[33]	2021	Undersampling technique with LR, DT, RF, KNN, SVM, and Naive Bayes	The implementation of various machine learning algorithms on the stroke dataset and the extension of training various models using neural networks were suggested. The work was limited to textual data, which might not always be accurate for stroke prediction
[34]	2023	Logistic regression and SVM with no resampling technique	The optimized logistic regression trained with tfidf transformed data showed the best performance for stroke prediction

TABLE 2: Description of the stroke dataset.

Attribute	Description	Unit	Data type
Gender	0 = Male 1 = Female	0, 1	Binary
Hypertension	0 = Nonhypertensive 1 = Hypertensive	0, 1	Binary
Heart diseases	0 = No 1 = Yes	0, 1	Binary
Ever married	0 = No 1 = Yes	0, 1	Binary
Work type	0 = Private 1 = Self-employed 2 = Government work 3 = Children	0, 1, 2, 3	Nominal
Residence type	0 = Urban 1 = Rural	0.1	Binary
Smoking status	0 = Never smoked 1 = Formerly smoked 2 = Smokes 3 = Unknown	0, 1, 2, 3	Nominal
Target (stroke)	0: No stroke 1: Stroke	0.1	Binary
Age	Age of an individual	In years	Numeric
Average glucose level	Glucose level of an individual	In Mmol/L	Numeric
BMI	Body mass index of an individual	In kg/m ²	Numeric

Figure 1(b) depicts the workflow of a study, starting with data collection and preprocessing, followed by data splitting. Here, the ML models are fitted using optimal hyperparameters obtained through grid search cross-validation and evaluated using the test set.

4. Mathematical Framework of the Study

In this section, the mathematical underpinning of the current study is provided.

4.1. KNNs Algorithm for Classification. The KNNs algorithm is a simple, yet powerful, nonparametric method used for classification and regression tasks. In the context of classifying stroke, KNN can be particularly effective due to its intuitive approach of leveraging the similarity between data points [37]. This section delves into the mathematical principles underlying the KNN algorithm, unraveling its application in stroke classification.

4.1.1. Distance Metric. The core idea of KNN is to classify a data point based on the majority class among its KNNs. The distance metric used to determine the “closeness” between data points is pivotal. The Euclidean distance, defined in (1), was employed in this study.

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}, \quad (1)$$

where p and q are two points in an n -dimensional feature space. The parameter k represents the number of nearest neighbors to consider for determining the class of a given

point. The choice of k is crucial, as a small k can lead to noise sensitivity, while a large k can cause the algorithm to overlook the local structure. A cross-validation technique was employed in this study to find a range of optimal k for the analysis of the data.

4.1.2. Classification Decision Rule. For a given point x to be classified, the steps are as follows:

- Compute the distance between x and all other points in the training set.
- Identify the k points in the training set that are closest to x .
- Assign the class label to x based on the majority vote among these KNNs. Formally, the class label $\hat{y}$ for the point x is given by the following equation:

$$\hat{y} = \arg \max_{c \in C} \sum_{i=1}^k 1(y_i = c), \quad (2)$$

where C is the set of all possible class labels, y_i is the class label of the i -th nearest neighbor, and $1(\cdot)$ is an indicator function that returns 1 if the condition is true, and 0 otherwise. The KNNs algorithm provides a straightforward yet effective method for stroke classification, leveraging the similarity between data points. Its nonparametric nature makes it versatile and easy to implement, though careful consideration must be given to the choice of distance metric, the value of k , and feature scaling. When appropriately tuned, KNN can yield valuable insights and accurate predictions in the domain of stroke diagnosis and prognosis.

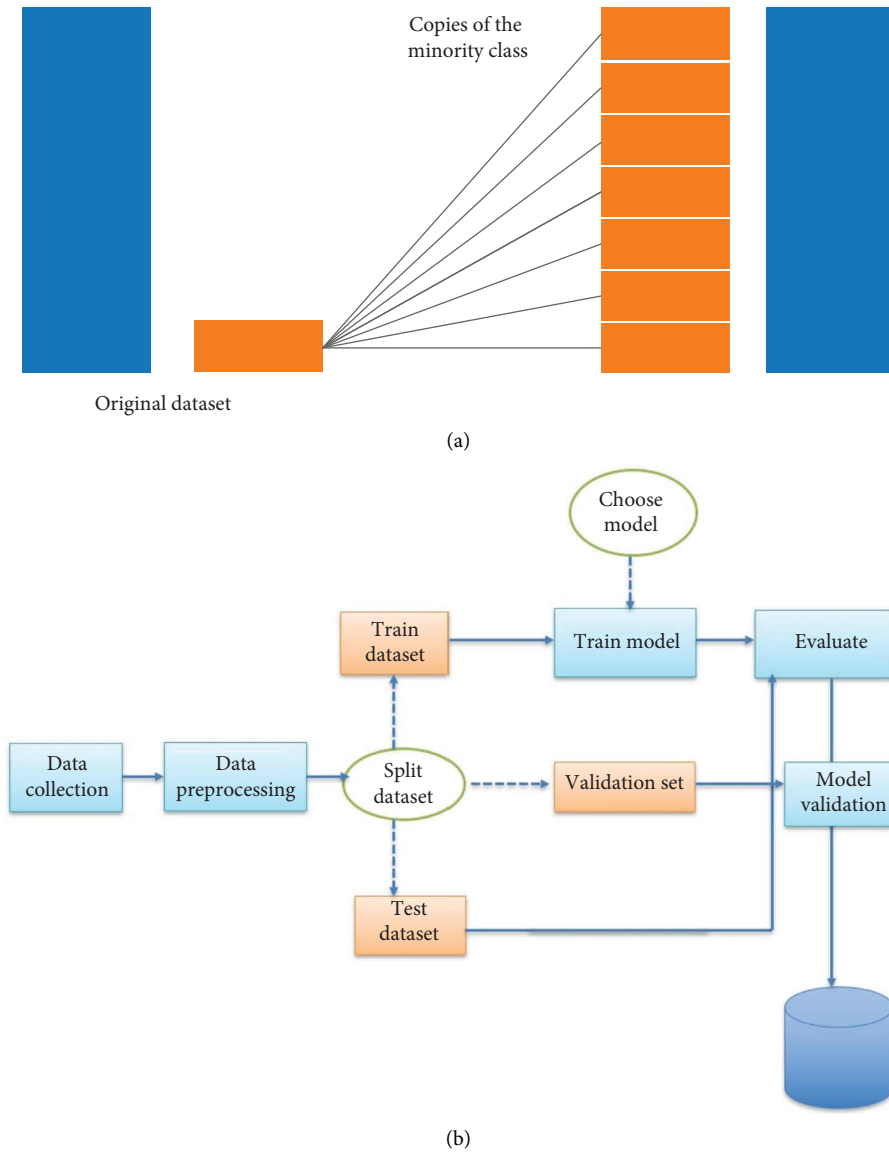


FIGURE 1: Working architecture and conceptual workflow adopted for the study. (a) Overview of oversampling techniques. (b) Workflow of the study.

4.2. RO. RO is a straightforward technique to handle a class imbalance in classification problems [38]. The method involves randomly duplicating samples from the minority class to balance the number of samples in each class. The steps for performing RO are as follows:

- i. Identify minority and majority class samples. Let $X_{\min}$ be the set of minority class samples and $X_{\max}$ be the set of majority class samples.
- ii. Determine the number of samples needed. Let $n_{\min}$ be the number of minority class samples and $n_{\max}$ be the number of majority class samples. The goal is to equalize the number of samples in both classes by increasing the number of minority samples to match the majority class.

iii. Calculate the number of duplicates needed. The number of additional samples needed to balance the classes is $n_{\max} - n_{\min}$.

iv. Select the random sample (with or without replacement) from $X_{\min}$ until the minority class size matches the majority class size.

4.3. SMOTE. SMOTE is an approach for addressing class imbalance in classification tasks. It generates synthetic samples for the minority class to balance the dataset [39]. The steps for performing SMOTE are as follows:

- i. Identify all the samples belonging to the minority class in the dataset. Let us denote the minority class samples as $X_{\min}$.

- ii. Decide how many synthetic samples are needed to balance the dataset. Suppose we need to generate N synthetic samples.
- iii. For each minority class sample $x_i \in X_{\min}$, find the KNNs ($NN[x_i]$) of x_i within the minority class. $NN(x_i) = \{x_{i1}, x_{i2}, \dots, x_{ik}\}$. Then, randomly select one of these neighbors, say x_{ij} .
- iv. Generate a new synthetic sample x_{smote} by interpolating between x_i and x_{ij} by using the following equation:

$$x_{\text{smote}} = x_i + \lambda \cdot (x_{ij} - x_i), \quad (3)$$

where λ is a random number between 0 and 1, i.e., $\lambda \sim \text{uniform}(0, 1)$. The interpolation step essentially creates a point along the line segment between x_i and x_{ij} . By varying λ between 0 and 1, the synthetic sample x_{smote} will lie somewhere between x_i and x_{ij} . This helps in creating a more diverse and distributed set of synthetic samples.

4.4. SMOTE-Tomek. SMOTE-Tomek is a hybrid technique that combines SMOTE with Tomek links to handle class imbalance and clean the data. This technique first applies SMOTE to generate synthetic samples and then uses Tomek links to remove borderline and noisy examples, thus improving the quality of the dataset [40]. Proceeding from SMOTE, the steps for performing SMOTE-Tomek are as follows:

- v. Identify Tomek links: A Tomek link exists between two samples x_i and x_j from different classes if they are each other's nearest neighbors: $d(x_i, x_j) < d(x_i, x_k)$ for any $x_k \neq x_j$, where d is the distance metric (typically Euclidean distance).
- vi. Remove all Tomek links to clean the dataset: This step involves removing either one or both samples from each Tomek link. A common practice employed in this study was to remove the majority class samples in the Tomek links to retain more minority class samples.

4.5. ADASYN. ADASYN is a technique for addressing class imbalance in classification problems. It extends SMOTE by not only generating synthetic samples but also focusing more on the samples that are harder to learn, i.e., those that are in sparsely populated regions [41]. The steps for performing ADASYN are as follows:

- i. Identify all the samples belonging to the minority class. Let the set of minority samples be $X_{\min}$.
- ii. Decide the total number of synthetic samples to generate, N .
- iii. For each minority sample x_i , find the KNNs ($NN[x_i]$) in the combined dataset (both minority and majority classes). Then, compute the ratio $r \in [0, 1]$ of majority class samples among the KNNs by using the following equation:

$$r_i = \frac{\text{number of majority class samples in } NN(x_i)}{k}. \quad (4)$$

- iv. Normalize the density ratios r_i to get $\hat{r}_i$ by using the following equation:

$$\hat{r}_i = \frac{r_i}{\sum_{i=1}^{n_{\min}} r_i}, \quad (5)$$

where $n_{\min}$ is the minority class sample.

- v. For each minority sample x_i , determine the number of synthetic samples G_i by using the following equation:

$$G_i = \hat{r}_i \times N. \quad (6)$$

For each synthetic sample to be generated, randomly select one of the KNNs x_{ij} . Generate a synthetic sample x_{adasyn} as

$$x_{\text{adasyn}} = x_i + \lambda \cdot (x_{ij} - x_i), \quad (7)$$

where $\lambda \sim \text{uniform}(0, 1)$.

The implementations of the algorithms used for the study are found in the Appendix in Section 6.

In this study, the model performance is measured based on the following parameters:

$$\text{precision} = \frac{TP}{TP + FP}, \quad (8)$$

$$\text{recall (sensitivity)} = \frac{TP}{TP + FN}, \quad (9)$$

$$F1 = \frac{2(\text{precision} \times \text{recall})}{\text{precision} + \text{recall}}, \quad (10)$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative. In addition, we employed the G-mean statistic given in (11), which is also an excellent indicator that effectively handles imbalanced class issues.

$$G - \text{mean} = \sqrt{\text{sensitivity} \times \text{specificity}}. \quad (11)$$

The G-mean is the geometric mean value used to measure the overall model performance. Poor classification results will produce a small G-mean value [42]. For details on the other ML algorithms adopted in this study such as SVM, logistic regression, RF, and DTs, see [43–45].

5. Results and Discussion

In this section, the results and discussions pertaining to the findings of the study are presented.

5.1. Descriptive Statistics of Research Variables. From Figure 2, the gender distribution indicates a higher number of females (2907) compared to males (2074). Regarding hypertension, most individuals do not have hypertension

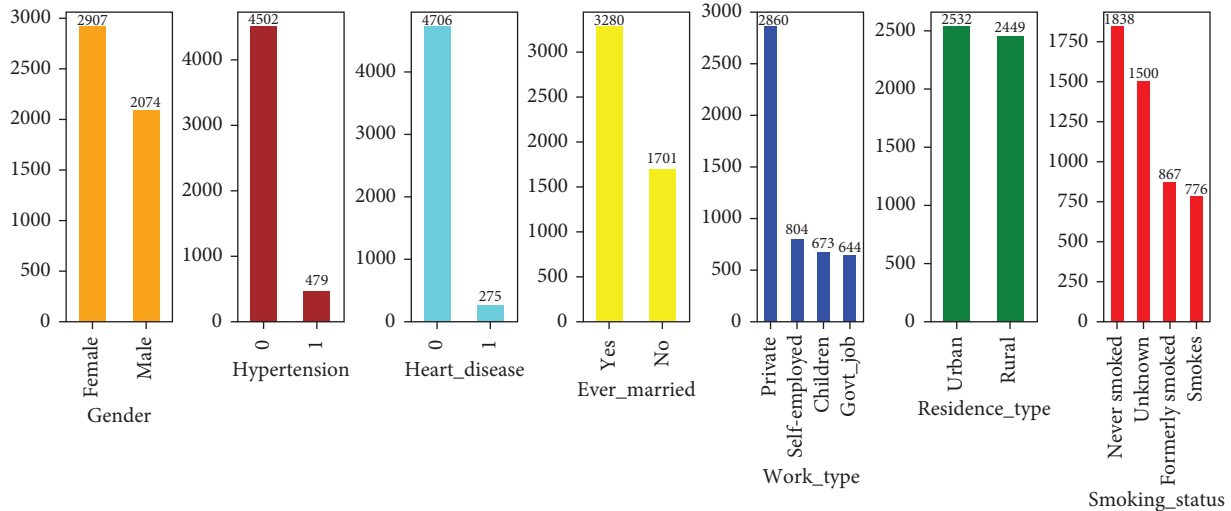


FIGURE 2: Bar charts showing the distribution of categorical variables.

(4502), while a smaller proportion are hypertensive (479). Similarly, the prevalence of heart disease is low, with 4706 individuals not having heart disease and only 275 being affected. Marital status shows that a significant portion of the dataset consists of individuals who have been married (3280), compared to those who have never married (1701). In terms of work type, most individuals are employed in the private sector (2860), followed by self-employed (804), children (673), and government jobs (644). The residence type distribution is nearly balanced, with 2532 individuals living in urban areas and 2449 in rural areas. Smoking status reveals that the largest group has never smoked (1838), followed by those with an unknown smoking status (1500), individuals who formerly smoked (867), and current smokers (776). These distributions provide valuable insights into the demographic and lifestyle characteristics of the population under study, which are essential for understanding the factors related to stroke classification.

From Figure 3, it is evident that age and average glucose level have a low positive correlation (0.24), suggesting a slight tendency for average glucose levels to increase with age. Age and body mass index (BMI) show a moderate positive correlation (0.37), indicating that BMI tends to be higher in older individuals. Average glucose level and BMI have a weak positive correlation (0.19), implying a slight relationship between these variables. Likewise, from Figure 3, the distribution of stroke cases shows a significant imbalance between the two categories. Specifically, there are 4733 instances of “no stroke” compared to only 248 instances of “stroke.” This stark imbalance of the stroke data highlights a common issue in medical datasets where the prevalence of a condition (stroke, in this case) is much lower than its absence. Such imbalanced data can pose significant challenges for classification algorithms, which may become biased towards the majority class (no stroke), leading to poor predictive performance for the minority class (stroke). To address this issue, it is crucial to consider oversampling techniques, such as RO, ADASYN, SMOTE, and SMOTE-Tomek to artificially balance the dataset.

5.2. RO, ADASYN, SMOTE, and SMOTE-Tomek Oversampling Techniques. From Figure 3, undersampling might not be a suitable technique for the stroke dataset under consideration because it would involve removing many instances from the majority class (no stroke), which is significantly larger than the minority class (stroke). This reduction could result in a loss of valuable information and lead to a model that does not generalize well.

On the other hand, oversampling the minority class (stroke) by creating synthetic instances can balance the dataset without discarding any existing data. This approach helps the model learn from more balanced data, potentially improving its ability to detect strokes accurately while maintaining all the critical information from the original dataset. Figure 4 presents the final samples after the application of the four oversampling techniques. As depicted in Figure 4(a), there is an equal number of “no stroke” and “stroke” instances achieved through RO, each with 3790 cases. Figure 4(b) represents the ADASYN technique, which balances the dataset by generating 3790 “no stroke” and 3849 “stroke” instances, demonstrating a slight increase in the number of synthetic samples for the minority class.

Figure 4(c) shows the result of the SMOTE, which creates an equal number of 3790 cases for both classes. Figure 4(d) depicts the SMOTE-Tomek technique, achieving a balanced dataset with 3774 instances in each class, which is indicative that 16 instances from Tomek links which were removed from SMOTE to generate the SMOTE-Tomek balanced the sample.

Figure 5 gives a detailed relationship between age, average glucose level, and BMI, with stroke status as the target. Each diagonal element of the plot shows the kernel density estimate (KDE) of the individual variables. The off-diagonal scatter plots depict pairwise relationships between these variables, with blue dots representing individuals who did not have a stroke (stroke = 0) and orange dots representing those who did (stroke = 1). The KDE plots reveal that most individuals in the dataset are in the middle age range, have moderate average glucose levels, and have a BMI clustered

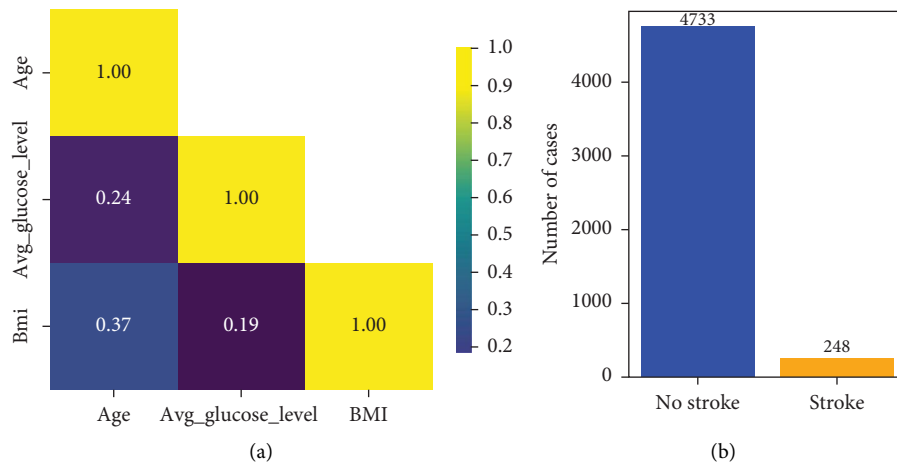


FIGURE 3: Correlogram of numerical variables and stroke count without oversampling. (a) Correlation matrix. (b) Stroke count without oversampling.

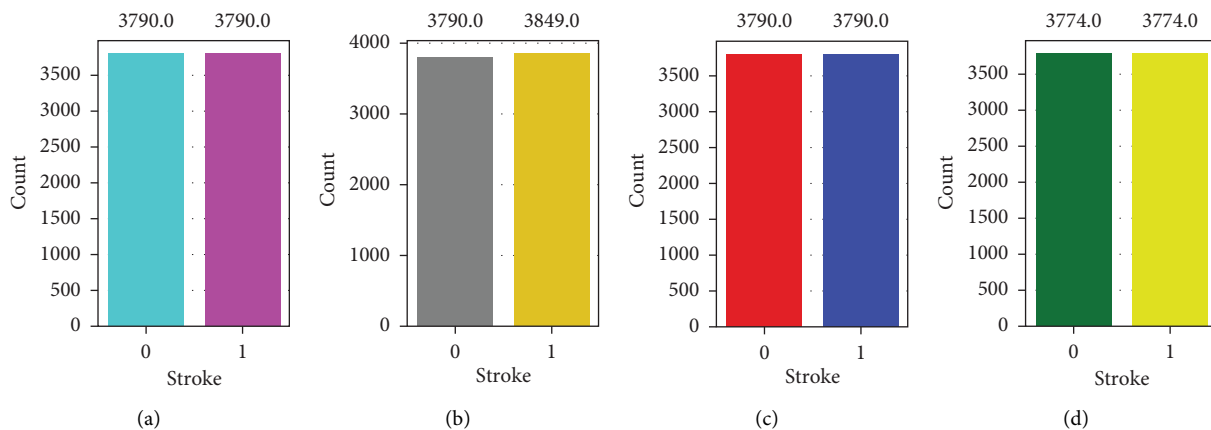


FIGURE 4: Bar charts showing stroke data counts after application of oversampling techniques. (a) RO. (b) ADASYN. (c) SMOTE. (d) SMOTE-Tomek.

around the mid-20–30 s. The scatter plots show some patterns where individuals with higher average glucose levels and higher ages are more likely to have had a stroke. In addition, there is a noticeable clustering of stroke cases at higher age ranges and higher BMI values, suggesting potential risk factors associated with stroke. Given the complex, nonlinear relationships between age, average glucose level, BMI, and stroke occurrence, it is indicative that stroke cases are not easily separable by a linear decision boundary. Despite this, this study adopted ML models such as KNN and SVM with RBF kernel with their nonlinear decision boundary which is particularly well-suited for this classification task.

5.3. Analysis of the KNN Algorithm Cross-Validation. An iteration over a range of k values from 1 to 30 was considered. For each value of k , a KNN classifier was created with the current k value. The KNN classifier was trained on the training data and predictions were made for the test data. The error rate for the current k was calculated as the mean of

the instances where the predicted labels do not match the true labels. After evaluating all k values, the error rates were paired with their corresponding k values and sorted in ascending order of error rates. This validation technique allows for an empirical selection of the optimal k value by minimizing the classification error on the test set, ensuring that the chosen k provides the best balance between bias and variance, leading to a more robust and accurate KNN model. For the KNN classifier, even though $k = 1, 3, 4, 5, 7, 9, 10, 11, 12, 13, 14$, and 15 were the twelve nearest neighbors with the least error rates out of the 30 k values considered, this study chose to analyze $k = 3, 5, 7, 9, 11, 13$, and 15 eliminating even numbered K values to avoid ties in the classification of stroke instances. Likewise, $k = 1$ was not chosen because it is so sensitive to just rely on 1 neighbor.

5.4. KNN Model Performance Evaluation. In this study, the classification was performed five times with five different data, namely, data without any oversampling (DWO) treatment (imbalanced data), data with RO, ADASYN,

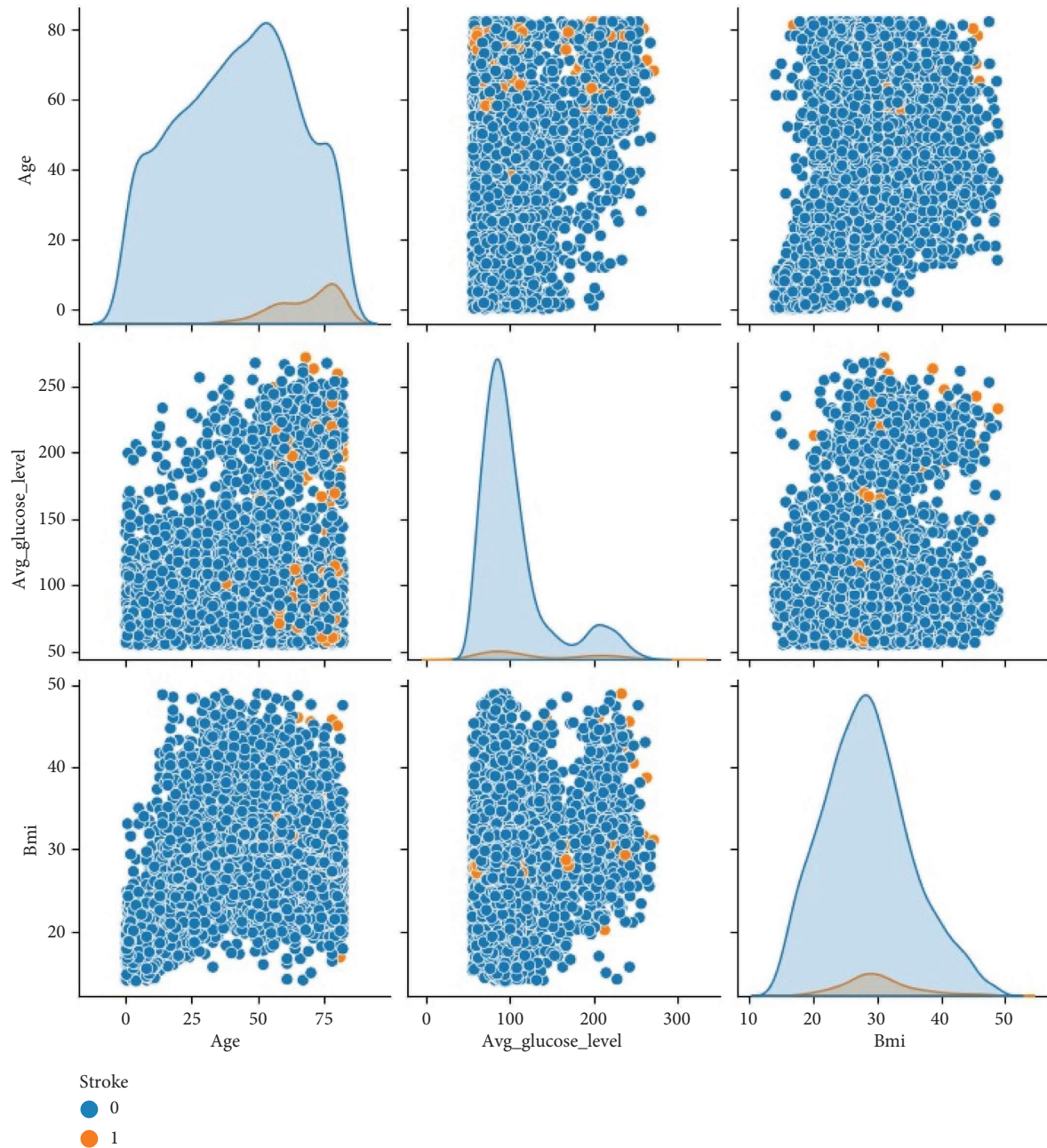


FIGURE 5: Bivariate relationships and univariate distributions.

SMOTE, and SMOTE-Tomek treatment using the KNN algorithm. Validation was carried out with various values of k (3, 5, 7, 9, 11, 13, and 15) based on the confusion matrix obtained. The results of the classification are presented in Table 3.

The study found out that validating the classification results solely based on accuracy was inadequate due to the imbalance between the “no stroke” and “stroke” classes. As shown in Table 3, the DWO treatment achieved high accuracy and specificity for all the values of k considered but exhibited very low sensitivity. This indicates that the model performed poorly in classifying the “stroke” class while being

highly effective in the classification of the “no stroke” class. Consequently, the DWO treatment was unsuitable for this classification task, as the results were predominantly influenced by the accuracy of the majority class (no stroke). In lieu of this, the authors in [46] recommended the usage of the G-mean statistic as a useful tool for evaluating classification models with imbalanced data. From Table 3, the G-mean values for data that were not oversampled were consistently low (0.0000) for k values of 3, 5, 7, 9, 11, 13, and 15 due to the sensitivity of all k values being zero. In contrast, using data treated with RO, ADASYN, SMOTE, and SMOTE-Tomek at $k = 3$ yielded the highest G-mean scores

TABLE 3: Classification results of oversampling techniques for different k values of KNN.

k	Data	Sensitivity	Specificity	G-mean	Precision	F1-score	Accuracy (%)
3	DWO	0.0000	0.9852	0.0000	0.0000	0.0000	93.18
	RO	0.8646	0.9088	0.8864	0.8977	0.8805	86.46
	ADASYN	0.8215	0.8579	0.8395	0.9008	0.8576	82.15
	SMOTE	0.8195	0.8547	0.8369	0.9020	0.8567	81.95
	SMOTE-Tomek	0.8205	0.8558	0.8379	0.9020	0.8573	82.05
5	DWO	0.0000	0.9989	0.0000	0.0000	0.0000	94.48
	RO	0.8375	0.8738	0.8555	0.9035	0.8677	83.75
	ADASYN	0.7854	0.8102	0.7977	0.9096	0.8378	78.54
	SMOTE	0.7914	0.8144	0.8028	0.9127	0.8421	79.14
	SMOTE-Tomek	0.7914	0.8134	0.8023	0.9140	0.8424	79.14
7	DWO	0.0000	0.9989	0.0000	0.0000	0.0000	94.48
	RO	0.8235	0.8462	0.8348	0.9179	0.8631	82.35
	ADASYN	0.7683	0.7847	0.7765	0.9175	0.8281	76.83
	SMOTE	0.7793	0.7953	0.7873	0.9196	0.8355	77.93
	SMOTE-Tomek	0.7783	0.7943	0.7863	0.9195	0.8349	77.83
9	DWO	0.0000	1.0000	0.0000	0.0000	0.0000	94.58
	RO	0.7964	0.8155	0.8059	0.9183	0.8463	79.64
	ADASYN	0.7472	0.7603	0.7537	0.9187	0.8143	74.72
	SMOTE	0.7543	0.7667	0.7605	0.9205	0.8192	75.43
	SMOTE-Tomek	0.7553	0.7678	0.7615	0.9206	0.8198	75.53
11	DWO	0.0000	1.0000	0.0000	0.0000	0.0000	94.58
	RO	0.7693	0.7847	0.7770	0.9189	0.8289	76.93
	ADASYN	0.7312	0.7391	0.7351	0.9231	0.8038	73.12
	SMOTE	0.7402	0.7487	0.7444	0.9237	0.8100	74.02
	SMOTE-Tomek	0.7412	0.7497	0.7454	0.9238	0.8107	74.12
13	DWO	0.0000	1.0000	0.0000	0.0000	0.0000	94.58
	RO	0.7442	0.7540	0.7491	0.9226	0.9226	77.42
	ADASYN	0.7131	0.7190	0.7160	0.9234	0.7912	71.31
	SMOTE	0.7302	0.7381	0.7341	0.9231	0.8031	73.02
	SMOTE-Tomek	0.7302	0.7381	0.7341	0.9231	0.8031	73.02
15	DWO	0.0000	1.0000	0.0000	0.0000	0.0000	94.58
	RO	0.7182	0.7253	0.7217	0.9223	0.7947	71.82
	ADASYN	0.7001	0.7041	0.7021	0.9241	0.7821	70.01
	SMOTE	0.7131	0.7190	0.7160	0.9234	0.7912	71.31
	SMOTE-Tomek	0.7131	0.7190	0.7160	0.9234	0.7912	71.31

of 0.8864, 0.8395, 0.8369, and 0.8379, respectively. This indicates that with balanced data, the resulting classifications for the stroke and no stroke classes were sufficiently accurate.

Figure 6 presents a graphical comparison of the four oversampling techniques using the testing data. The study found the RO technique to be superior compared to the other three oversampling techniques (ADASYN, SMOTE, and SMOTE-Tomek) after the consideration of all the various performance evaluation metrics including sensitivity, specificity, G-mean, F1-score and accuracy scores. From Figure 6, the comparison of the four oversampling techniques, RO, ADASYN, SMOTE, and SMOTE-Tomek, for $k=3$ highlights the distinct performance characteristics across G-mean, F1-score, and accuracy metrics. RO demonstrates superior performance with the highest G-mean (0.8864), F1-score (0.8805), and a notable accuracy of 86.46%, indicating its robustness in balancing sensitivity (0.8646) and specificity (0.9088) as well as precision (0.8977).

ADASYN also shows strong results with a G-mean of 0.8395, an F1-score of 0.8576, and an accuracy of 82.15%, making it a reliable method albeit slightly less effective than RO. SMOTE-Tomek provides comparable performance to SMOTE, with a G-mean of 0.8379, an F1-score of 0.8573, and an accuracy of 82.05%, reflecting a marginal trade-off in balance. SMOTE, while achieving a high F1-score of 0.8567, has a slightly lower G-mean (0.8369) and the lowest accuracy at 81.95%, indicating that it is less effective in maintaining overall performance balance compared to the other oversampling methods. Overall, $k=3$ with RO stands out as the most effective oversampling technique in terms of G-mean, F1-score, and accuracy.

The findings of this present study are consistent with other studies in literature, such as [38], which focuses on the implementation of the KNN algorithm using oversampling techniques, highlighting the importance of using oversampling techniques to enhance the accuracy and reliability of classification models in addressing socioeconomic

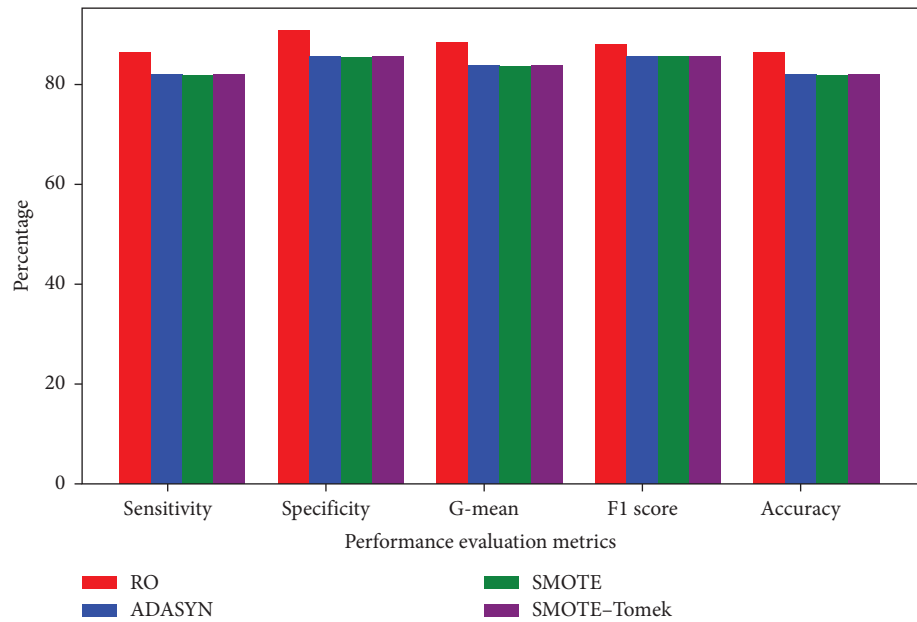


FIGURE 6: Comparison of the KNN model based on performance evaluation metrics for $k = 3$.

TABLE 4: Other machine learning classification results of oversampling techniques.

Model	Data	Sensitivity	Specificity	G-mean	Precision	F1-score	Accuracy (%)
SVM (RBF)	DWO	0.0000	1.0000	0.0000	0.0000	0.0000	94.58
	RO	0.8987	0.9491	0.9236	0.8941	0.8964	89.87
	ADASYN	0.8806	0.9268	0.9034	0.8976	0.8890	88.06
	SMOTE	0.8806	0.9290	0.9045	0.8944	0.8874	88.06
	SMOTE-Tomek	0.8776	0.9258	0.9014	0.8942	0.8858	87.76
SVM (linear)	DWO	0.9458	0.0000	0.0000	0.0000	0.0000	94.58
	RO	0.7202	0.7116	0.7159	0.9441	0.7967	72.02
	ADASYN	0.7272	0.7190	0.7231	0.9443	0.8017	72.72
	SMOTE	0.7302	0.9290	0.8236	0.9430	0.8038	73.02
	SMOTE-Tomek	0.7302	0.7232	0.7267	0.9430	0.8038	73.02
Logistic regression	DWO	0.0000	0.9979	0.0000	0.0000	0.0000	99.79
	RO	0.7362	0.7285	0.7323	0.9447	0.8081	73.62
	ADASYN	0.7392	0.7328	0.7360	0.9434	0.8102	73.92
	SMOTE	0.7482	0.7423	0.7452	0.9438	0.8166	74.82
	SMOTE-Tomek	0.7482	0.7423	0.7452	0.9438	0.8166	74.82
RF	DWO	0.0000	0.9968	0.0000	0.0000	0.0000	94.28
	RO	0.7292	0.7222	0.7257	0.9429	0.8031	72.92
	ADASYN	0.7372	0.7381	0.7376	0.9333	0.8084	73.72
	SMOTE	0.7432	0.7423	0.7427	0.9364	0.8128	74.32
	SMOTE-Tomek	0.7472	0.7466	0.7469	0.9366	0.8155	74.72
DT	DWO	0.0000	0.9523	0.0000	0.0000	0.0000	91.78
	RO	0.7633	0.7646	0.7639	0.9361	0.8265	76.33
	ADASYN	0.7924	0.8038	0.7981	0.9271	0.8450	79.24
	SMOTE	0.7653	0.7699	0.7676	0.9321	0.8276	76.53
	SMOTE-Tomek	0.7884	0.7943	0.7913	0.9335	0.8431	78.84

Abbreviation: DWO, data without oversampling.

TABLE 5: Comparison of our proposed method with existing similar works.

Research work, year	Best algorithm	Dataset used	Highest metrics
[31], 2024	Hybrid neural network and random forest (NN-RF) via adaptive synthetic sampling oversampling (ADASYN)	Stroke dataset from Kaggle	84.00% accuracy and an F1-score of 75.00%
[23], 2022	RegNetY-320 with a learning rate of 0.001	HAM10000 dataset	91% accuracy and an F1-score of 88.1%
[49], 2022	Decision trees (D/Ts) induced via genetic programming (GP)	Stroke dataset from Kaggle	Accuracy of 70.00%
[50], 2019	Automated hyperparameter optimization (AutoHPO) based on DNN	Stroke dataset from Kaggle	Accuracy of 72.80%
[51], 2022	Collective intelligence-based system	HAM10000 dataset	Accuracy of 86.70%
[52], 2022	Wide-ShuffleNet	HAM10000 dataset	Accuracy of 86.30%
[53], 2021	Mask-RCNN	HAM10000 dataset	86.50% accuracy and an F1-score of 86.20%
[54], 2021	DenseNet-201	HAM10000 dataset	87.70% accuracy and an F1-score of 85.50%
[55], 2021	CRCKD algorithm	HAM10000 dataset	85.60% accuracy and an F1-score of 76.40%
[56], 2021	MobileNet	HAM10000 dataset	83.10% accuracy and an F1-score of 83.00%
[57], 2022	DenseNet-201	HAM10000 Dataset	82.90% accuracy and an F1-score of 74.40%
[32], 2022	NN with random downsampling	Stroke dataset Prediction	Accuracy of 77.00% and an F1-score of 77.00%
[33], 2021	Naive Bayes coupled with undersampling	Stroke dataset Prediction	Accuracy of 82.00% and an F1-score of 82.30%
[34], 2023	SVM with no resampling technique	EHR (MIMIC-III)	Accuracy of 73.00%
[58], 2021	Random forest via undersampling-clus tering-oversampling	Stroke samples	Accuracy of 70.37% and an F1-score of 75.32%
Proposed method	SVM with RBF kernel in combination with random oversampling	Stroke dataset from Kaggle	89.87% accuracy and an F1-score of 89.64%

disparities related to household welfare status. Also, the studies in [47, 48] highlight the importance of addressing the class imbalance problem using oversampling approaches such as SMOTE, ADASYN, KNN oversampling (KNNOR), and sigma nearest oversampling based on convex combination (SNOCC) for enhanced sensitivity, specificity, and precision.

5.5. Classification Results of Oversampling Techniques for Other ML Algorithms. The comparative analysis of the other four different ML algorithms, SVM, logistic regression, RF, and DTs, in combination with various oversampling techniques is presented in Table 4.

The key metrics evaluated in Table 4 include sensitivity, specificity, G-mean, precision, F1-score, and accuracy. Starting with the SVM with an RBF kernel, the DWO performed poorly, particularly in terms of sensitivity and G-mean. This indicates the model's inability to effectively detect the minority class, "stroke." However, when RO was applied, the model's performance significantly improved, achieving the highest G-mean of 0.9236. This improvement reflects a good balance between sensitivity (0.8987) and specificity (0.9491). Both ADASYN and SMOTE also showed similar improvements, though with slight variations in G-mean and F1-score, indicating their effectiveness in addressing class imbalance. It was observed that RO-SVM, ADASYN-SVM, SMOTE-SVM, and SMOTE + Tomek-SVM all performed very well on the stroke data compared to RO-KNN with $k=3$. The SVM with a linear kernel demonstrated less effectiveness compared to the RBF kernel, particularly in handling the imbalanced data. Despite this, the application of oversampling techniques such as RO, ADASYN, SMOTE, and SMOTE-Tomek still resulted in moderate performance enhancements, with G-mean values ranging from 0.7159 to 0.8236. This suggests that while the linear kernel can benefit from oversampling, it may not be as robust as the RBF kernel in this context. Logistic regression, much like the SVM with a linear kernel, struggled with the imbalanced dataset in its DWO form. However, the introduction of oversampling techniques significantly improved its performance. Among these, SMOTE and SMOTE-Tomek emerged as the most effective, achieving the highest equivalent G-mean of 0.7452. RF models also showed considerable improvement when combined with oversampling techniques. SMOTE-Tomek led to the best performance, with a G-mean of 0.7469, indicating a well-balanced sensitivity and specificity. Finally, DTs demonstrated a strong response to oversampling, especially when paired with ADASYN, which achieved the highest G-mean of 0.7981. RO and SMOTE-Tomek also performed well, further indicating that DTs, while simple, can greatly benefit from techniques that balance the dataset. These analyses indicate that oversampling techniques, particularly RO and SMOTE-Tomek, play a critical role in enhancing the

performance of various ML algorithms when dealing with imbalanced datasets.

5.6. Comparative Analysis With State-of-the-Art Techniques. In this section, a comparison of the proposed SVM with RBF kernel via RO technique is performed with the state-of-the-art studies to validate its efficacy and robustness. The best hybrid method in this study demonstrated promising outcomes across various evaluation metrics in the domain of stroke disease prediction. Comparative analyses with numerous contemporary studies, considering the methods applied, datasets used, and accuracy and F1-scores achieved, were conducted, as shown in Table 5. Specifically, the proposed framework in this study achieved an accuracy and F1-score of 89.87% and 89.64%, respectively, indicating its competitiveness with the state-of-the-art techniques.

6. Conclusion and Recommendations

Class imbalance is a significant challenge when working with health data. In many medical datasets, the distribution of classes can be highly skewed. For example, datasets related to disease diagnosis often contain a majority of nondisease cases and a minority of disease cases just as was evident with the stroke data considered in this study. This imbalance can cause ML models to perform poorly, as they may become biased towards the majority class and fail to accurately predict the minority class, which is often the class of greater interest in healthcare applications. Addressing class imbalance in health data is essential for developing accurate and reliable ML models. Techniques such as RO, ADASYN, SMOTE, and SMOTE-Tomek provide effective ways to balance datasets, thereby improving model performance and ensuring fair predictions. In this study, the challenge of class imbalance in health data classification, focusing on the application of various oversampling techniques to improve the performance of KNNs, SVM, logistic regression, RF, and DTs for stroke prediction, is explored. The analysis confirmed that the presence of a significant class imbalance in the dataset, with a much higher number of "no stroke" cases compared to "stroke" cases, can lead to biased models that fail to accurately predict minority class instances. To address this, the study applied four different oversampling techniques: RO, ADASYN, SMOTE, and SMOTE-Tomek.

This study has significantly contributed to the field of health data classification with a focus on stroke prediction. Initially, the impact of several oversampling techniques, RO, ADASYN, SMOTE, and SMOTE-Tomek, on enhancing ML model's performance was evaluated. The findings of the study indicated that all oversampling techniques applied in conjunction with the ML models adopted for the study improved the model's ability to classify stroke cases compared to the imbalanced dataset. Among these, RO in conjunction with SVM-RBF kernel showed promising results, achieving the

best performance in terms of sensitivity, specificity, G-mean, F1-score, and accuracy of 89.87%, 94.91%, 92.36%, 89.64%, and 89.87%, respectively. Furthermore, an exhaustive comparison of various ML algorithms, such as KNNs, SVM, logistic regression, RF, and DTs, is provided. This analysis identified optimal combinations of models and oversampling techniques that yielded the best outcomes for predicting stroke. For instance, the study discovered that while all oversampling methods improved classification accuracy, specific configurations such as the RO-SVM stood out for their superior performance metrics. RO also showed promising results for the KNN model, providing the highest G-mean, F1-score, and accuracy, thus balancing the sensitivity and specificity effectively for all k values under consideration. Likewise, for the KNN model, ADASYN, SMOTE, and SMOTE-Tomek also showed substantial improvements, with ADASYN performing slightly better than SMOTE and SMOTE-Tomek in most performance metrics considered. The study highlights the importance of addressing the class imbalance in health datasets to build reliable and fair predictive models. By enhancing the representation of the minority class through oversampling, models that not only achieve higher accuracy but also ensure better sensitivity and recall, crucial for early and accurate disease diagnosis, can be developed. These improvements are essential for making informed clinical decisions and improving patient outcomes. Implementing these techniques can significantly enhance the ability of classification models to correctly identify minority class instances, ultimately leading to better healthcare outcomes.

Each oversampling technique adopted in this study to address class imbalance has its own strengths and limitations. For RO, a key limitation is the risk of overfitting, as duplicating existing minority class samples can cause the model to become too tailored to the training data, resulting in poor generalization to new data [59]. The authors in [60] highlighted that RO, while effective in balancing datasets, often leads to models that perform well on training data but poorly on unseen data due to overfitting. RO is most effective in scenarios where the model's variance is a significant issue, but its simplicity can be a limitation in complex datasets. ADASYN, while focusing on generating more synthetic samples in regions with higher learning difficulty, can sometimes create synthetic samples that are too similar to each other or to the majority class, leading to less effective improvement in model performance [61]. The authors in [41] showed that ADASYN often results in better classification performance than SMOTE, particularly in highly imbalanced datasets, by emphasizing the areas of the feature space where the classifier struggles the most. This targeted approach helps in creating a more robust model that generalizes better across different data distributions. However, the authors in [62] noted that ADASYN can sometimes produce synthetic samples that are too similar to the majority class, which can

diminish its effectiveness in certain scenarios. SMOTE can create samples that lie too close to the majority class boundary, potentially increasing the chance of misclassification [63]. Finally, SMOTE-Tomek, which combines SMOTE with Tomek links to clean the dataset by removing noisy samples, may inadvertently remove useful borderline samples along with noisy ones, potentially leading to a loss of valuable information that could help the model better distinguish between classes. The authors in [64] demonstrated that this hybrid method further improves classification accuracy by refining the decision boundaries and reducing noise, making it particularly useful in datasets where class overlap is an issue. The removal of Tomek links enhances the dataset's clarity and the classifier's ability to learn distinct patterns. These oversampling techniques can be applied to other health-related diseases such as COVID-19 [65].

The authors in [66] evaluated undersampling, oversampling, a combination of oversampling and undersampling, and ensemble learning methods across various datasets and concluded that SMOTE-Tomek often provided the best balance between improving minority class recall and maintaining overall accuracy, especially in medical datasets. The study found that while ADASYN excelled in highly imbalanced datasets, SMOTE-Tomek offered more consistent improvements across different levels of imbalance. To conclude, the choice of oversampling method should be based on the specific characteristics of the dataset, as the authors in [20] noted that while some methods such as random oversampling (RO) and cost-sensitive algorithms show promise, there are inconsistencies and gaps in the current research. For instance, SMOTE-Tomek is ideal for datasets with significant class overlap, while ADASYN may be more suitable for cases with severe class imbalance. RO can be useful for simpler datasets where model variance is the primary concern, but it is generally outperformed by more sophisticated methods such as SMOTE and ADASYN in complex datasets. In lieu of these limitations, future research could explore the integration of deep learning techniques with traditional ML models, which offers a promising avenue for further enhancing the performance of health data classification systems. Deep learning ability to handle vast amounts of data and learn complex, abstract features could be leveraged to develop more robust models for stroke prediction. This approach could enable the discovery of intricate patterns in medical datasets that are not readily captured by current methods. Future research should explore the synergistic effects of these methods to refine predictive accuracy and reliability, ultimately leading to improved diagnostic processes and healthcare outcomes as noted by the authors in [23, 67].

Appendix A:


```

### Algorithm for Implementing SMOTE
for each sample  $x_i$  in  $X_{min}$ :
  find k-nearest neighbors of  $x_i$ 
  for  $j$  in range( $N / \text{len}(X_{min})$ ):
    randomly select one of the k-nearest neighbors  $x_{ij}$ 
    generate synthetic sample  $x_{new}$  using  $x_{new} = x_i + \lambda(x_{ij} - x_i)$ 
    add  $x_{new}$  to  $X_{new}$ 

```

```

### Algorithm for Implementing SMOTE-Tomek
Proceeding from SMOTE;
1. Combine  $X_{min}$ ,  $X_{maj}$ , and  $X_{new}$  to form  $X_{combined}$ .
2. Identify Tomek links in  $X_{combined}$ :
  for each pair  $(x_i, x_j)$  where  $x_i$  belongs to one class and  $x_j$  to another:
    if  $x_i$  and  $x_j$  are each others nearest neighbors, mark the pair as a Tomek link
3. Remove Tomek links from  $X_{combined}$ :
  for each Tomek link  $(x_i, x_j)$ :
    remove the majority class sample (or both samples) from the dataset

```

```

### Algorithm for Implementing ADASYN
for each sample  $x_i$  in  $X_{min}$ :
  find k-nearest neighbors of  $x_i$ 
  compute the ratio  $r_i$  of majority class samples in the k-nearest neighbors
  normalize the density ratios  $r_i$  to get  $\hat{r}_i$ 
  determine the number of synthetic samples  $G_i = \hat{r}_i * N$ 
  for  $j$  in range( $G_i$ ):
    randomly select one of the k-nearest neighbors  $x_{ij}$ 
    generate synthetic sample  $x_{new}$  using  $x_{new} = x_i + \lambda(x_{ij} - x_i)$ 
    add  $x_{new}$  to  $X_{new}$ 

```

```

### Algorithm for Implementing RO
for each sample  $x_i$  in  $X_{min}$ :
  determine the number of duplicates needed  $D$  such that the total samples in the
    minority class match the majority class size.
  for  $j$  in range  $D$ :
    randomly select  $x_i$  from  $X_{min}$ 
  Add  $x_i$  to  $X_{new}$ 

```

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Edmund Fosu Agyemang: conceptualization, methodology, writing—original draft, visualization, writing, reviewing, editing, project administration, methodology, formal analysis, supervision, resources, investigation, and validation.

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References

- [1] C. Victoria Ibeh, O. Adijat Elufioye, T. Olorunsogo, O. Franca Asuzu, N. Leonard Nduubuisi, and A. I. Daraojimba, "Data Analytics in Healthcare: A Review of Patient-Centric Approaches and Healthcare Delivery," *World Journal of Advanced Research and Reviews* 21, no. 2 (2024): 1750–1760, <https://doi.org/10.30574/wjarr.2024.21.2.0246>.
- [2] N. Budhathoki, R. Bhandari, S. Bashyal, and C. Lee, "Predicting Asthma Using Imbalanced Data Modeling Techniques: Evidence From 2019 Michigan Brfss Data," *Plos One* 18, no. 12 (2023): e0295427, <https://doi.org/10.1371/journal.pone.0295427>.
- [3] Y. Kim, W. Choi, W. Choi, et al., "A Machine Learning Approach Using Conditional Normalizing Flow to Address Extreme Class Imbalance Problems in Personal Health Records," *BioData Mining* 17, no. 1 (2024): 14–18, <https://doi.org/10.1186/s13040-024-00366-0>.
- [4] S. Feng, J. Keung, X. Yu, et al., "Coste: Complexity-Based Oversampling Technique to Alleviate the Class Imbalance Problem in Software Defect Prediction," *Information and Software Technology* 129 (2021): 106432, <https://doi.org/10.1016/j.infsof.2020.106432>.
- [5] M. S. Shelke, P. R. Deshmukh, and V. K. Shandilya, "A Review on Imbalanced Data Handling Using Undersampling and Oversampling Technique," *International Journal of Engineering Research in Current Trends* 3, no. 4 (2017): 444–449.
- [6] P. Soltanzadeh, M. R. Feizi-Derakhshi, and M. Hashemzadeh, "Addressing the Class-Imbalance and Class-Overlap Problems by a Metaheuristic-Based Under-Sampling Approach," *Pattern Recognition* 143 (2023): 109721, <https://doi.org/10.1016/j.patcog.2023.109721>.
- [7] T. Wongvorachan, S. He, and O. Bulut, "A Comparison of Undersampling, Oversampling, and Smote Methods for Dealing With Imbalanced Classification in Educational Data Mining," *Information* 14, no. 1 (2023): 54, <https://doi.org/10.3390/info14010054>.
- [8] C. Vairetti, J. L. Assadi, and S. Maldonado, "Efficient Hybrid Oversampling and Intelligent Undersampling for Imbalanced Big Data Classification," *Expert Systems With Applications* 246 (2024): 123149, <https://doi.org/10.1016/j.eswa.2024.123149>.
- [9] Z. Sun, W. Ying, W. Zhang, and S. Gong, "Undersampling Method Based on Minority Class Density for Imbalanced Data," *Expert Systems With Applications* 249 (2024): 123328, <https://doi.org/10.1016/j.eswa.2024.123328>.
- [10] S. Iqbal, A. N. Qureshi, J. Li, I. A. Choudhry, and T. Mahmood, "Dynamic Learning for Imbalanced Data in Learning Chest X-Ray and Ct Images," *Heliyon* 9, no. 6 (2023): e16807, <https://doi.org/10.1016/j.heliyon.2023.e16807>.
- [11] M. Al Moteri, T. R. Mahesh, A. Thakur, V. Vinoth Kumar, S. B. Khan, and M. Alojail, "Enhancing Accessibility for Improved Diagnosis With Modified Efficientnetv2-S and Cyclic Learning Rate Strategy in Women With Disabilities and Breast Cancer," *Frontiers of Medicine* 11 (2024): 1373244, <https://doi.org/10.3389/fmed.2024.1373244>.
- [12] J. M. Johnson and T. M. Khoshgoftaar, "Survey on Deep Learning With Class Imbalance," *Journal of Big Data* 6, no. 1 (2019): 27–54, <https://doi.org/10.1186/s40537-019-0192-5>.
- [13] A. Ghavidel and P. Pazos, "Machine Learning (ML) Techniques to Predict Breast Cancer in Imbalanced Datasets: A Systematic Review," *Journal of Cancer Survivorship: Research and Practice* (2023): 1–25, <https://doi.org/10.1007/s11764-023-01465-3>.
- [14] E. F. Agyemang, "Anomaly Detection Using Unsupervised Machine Learning Algorithms: A Simulation Study," *Scientific African* 26 (2024): e02386, <https://doi.org/10.1016/j.sciaf.2024.e02386>.
- [15] E. N. Noi Nortey, E. Fosu Agyemang, R. Minkah, and K. Asah-Asante, "Bayesian Estimation of Presidential Elections in Ghana: A Validation Approach," *African Journal of Applied Statistics* 9, no. 1 (2022): 1297–1317.
- [16] E. F. Agyemang, E. N. Nortey, R. Minkah, and K. Asah-Asante, "The Unfolding Mystery of the Numbers: First and Second Digits Based Comparative Tests and Its Application to Ghana's Elections," *Model Assisted Statistics and Applications* 18, no. 2 (2023): 183–192, <https://doi.org/10.3233/mas-221418>.
- [17] E. F. Agyemang, E. N. N. Nortey, R. Minkah, and K. Asah-Asante, "Baseline Comparative Analysis and Review of Election Forensics: Application to Ghana's 2012 and 2020 Presidential Elections," *Heliyon* 9, no. 8 (2023): e18276, <https://doi.org/10.1016/j.heliyon.2023.e18276>.
- [18] S. Rezvani and X. Wang, "A Broad Review on Class Imbalance Learning Techniques," *Applied Soft Computing* 143 (2023): 110415, <https://doi.org/10.1016/j.asoc.2023.110415>.
- [19] P. Kaur and A. Gosain, "Robust Hybrid Data-Level Sampling Approach to Handle Imbalanced Data during Classification," *Soft Computing* 24, no. 20 (2020): 15715–15732, <https://doi.org/10.1007/s00500-020-04901-z>.
- [20] J. L. Leevy, T. M. Khoshgoftaar, R. A. Bauder, and N. Seliya, "A Survey on Addressing High-Class Imbalance in Big Data," *Journal of Big Data* 5, no. 1 (2018): 42–30, <https://doi.org/10.1186/s40537-018-0151-6>.
- [21] L. Devnath, S. Luo, P. Summons, et al., "Deep Ensemble Learning for the Automatic Detection of Pneumoconiosis in Coal Worker's Chest X-Ray Radiography," *Journal of Clinical Medicine* 11, no. 18 (2022): 5342, <https://doi.org/10.3390/jcm11185342>.
- [22] X. Yang, M. Khushi, and K. Shaukat, "Biomarker Ca125 Feature Engineering and Class Imbalance Learning Improves Ovarian Cancer Prediction," in *2020 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE) (IEEE, 2020)*, 1–6.
- [23] T. M. Alam, K. Shaukat, W. A. Khan, et al., "An Efficient Deep Learning-Based Skin Cancer Classifier for an Imbalanced Dataset," *Diagnostics* 12, no. 9 (2022): 2115, <https://doi.org/10.3390/diagnostics12092115>.
- [24] M. Khushi, K. Shaukat, T. M. Alam, et al., "A Comparative Performance Analysis of Data Resampling Methods on Imbalance Medical Data," *IEEE Access* 9 (2021): 109960–109975, <https://doi.org/10.1109/access.2021.3102399>.
- [25] M. G. Alsubaiea, S. Luo, and K. Shaukat, "Convadd: Exploring a Novel Cnn Architecture for Alzheimer's Disease Detection," *International Journal of Advanced Computer Science and Applications* 15, no. 4 (2024): <https://doi.org/10.14569/ijacsa.2024.0150431>.
- [26] C. Srinivas, K. S. Nandini Prasad, M. Zakariah, et al., "Deep Transfer Learning Approaches in Performance Analysis of Brain Tumor Classification Using Mri Images," *Journal of Healthcare Engineering* 2022, no. 1 (2022): 3264367–3272022, <https://doi.org/10.1155/2022/3264367>.

- [27] M. R. Kumar, S. Vekkot, S. Lalitha, et al., "Dementia Detection From Speech Using Machine Learning and Deep Learning Architectures," *Sensors* 22, no. 23 (2022): 9311, <https://doi.org/10.3390/s22239311>.
- [28] R. Mia, S. Khanam, A. Mahjabeen, et al., "Exploring Machine Learning for Predicting Cerebral Stroke: A Study in Discovery," *Electronics* 13, no. 4 (2024): 686, <https://doi.org/10.3390/electronics13040686>.
- [29] E. Dritsas and M. Trigka, "Stroke Risk Prediction With Machine Learning Techniques," *Sensors* 22, no. 13 (2022): 4670, <https://doi.org/10.3390/s22134670>.
- [30] M. Abdullah Aish, A. Abdul Ghafoor, F. Nasim, K. I. Ali, S. Akhter, and S. Azeem, "Improving Stroke Prediction Accuracy Through Machine Learning and Synthetic Minority Over-Sampling," *Journal of Computing and Biomedical Informatics* 7, no. 02 (2024).
- [31] V. S. Elangovan, R. Devarajan, O. I. Khalaf, M. S. Sharif, and W. Elmedany, "Analysing an Imbalanced Stroke Prediction Dataset Using Machine Learning Techniques," *Karbala International Journal of Modern Science* 10, no. 2 (2024): 8, <https://doi.org/10.33640/2405-609x.3355>.
- [32] S. Dev, H. Wang, C. S. Nwosu, N. Jain, B. Veeravalli, and D. John, "A Predictive Analytics Approach for Stroke Prediction Using Machine Learning and Neural Networks," *Healthcare Analytics* 2 (2022): 100032, <https://doi.org/10.1016/j.health.2022.100032>.
- [33] G. Sailasya and G. L. A. Kumari, "Analyzing the Performance of Stroke Prediction Using ML Classification Algorithms," *International Journal of Advanced Computer Science and Applications* 12, no. 6 (2021): <https://doi.org/10.14569/ijacsa.2021.0120662>.
- [34] S. Jiang, Y. Gu, and E. Kumar, "Stroke Risk Prediction Using Artificial Intelligence Techniques Through Electronic Health Records," *Artificial Intelligence Evolution* (2023): 88–98, <https://doi.org/10.37256/aie.4120232744>.
- [35] J. Gou, H. Ma, W. Ou, S. Zeng, Y. Rao, and H. Yang, "A Generalized Mean Distance-Based K-Nearest Neighbor Classifier," *Expert Systems With Applications* 115 (2019): 356–372, <https://doi.org/10.1016/j.eswa.2018.08.021>.
- [36] E. F. Agyemang, J. A. Mensah, O.-A. Ampomah, et al., "Predicting Students' Academic Performance via Machine Learning Algorithms: An Empirical Review and Practical Application" (2024).
- [37] V. P. Prasetyo, M. F. A. U. Nuha, M. H. Hakiki, et al., "Comparison of Data Mining Techniques on Stroke Clinical Dataset," *Procedia Computer Science* 234 (2024): 502–511, <https://doi.org/10.1016/j.procs.2024.03.033>.
- [38] N. M. Djafar and A. Fauzan, "Implementation of K-Nearest Neighbor Using the Oversampling Technique on Mixed Data for the Classification of Household Welfare Status," *Statistics in Transition New Series* 25, no. 1 (2024): 109–124, <https://doi.org/10.59170/stattrans-2024-007>.
- [39] A. J. Mohammed, M. M. Hassan, and D. Hussein Kadir, "Improving Classification Performance for a Novel Imbalanced Medical Dataset Using Smote Method," *International Journal of Advanced Trends in Computer Science and Engineering* 9, no. 3 (2020): 3161–3172, <https://doi.org/10.30534/ijatcse/2020/104932020>.
- [40] H. Hairani, A. Anggrawan, and D. Priyanto, "Improvement Performance of the Random Forest Method on Unbalanced Diabetes Data Classification Using Smote-Tomek Link," *JOIV: International Journal on Informatics Visualization* 7, no. 1 (2023): 258–264, <https://doi.org/10.30630/joiv.7.1.1069>.
- [41] H. He, Y. Bai, E. A. Garcia, and S. Li, "Adasyn: Adaptive Synthetic Sampling Approach for Imbalanced Learning," in *2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)* (IEEE, 2008), 1322–1328.
- [42] M. Bekkar, T. A. Alitouche, and T. Akrouf Alitouche, "Imbalanced Data Learning Approaches Review," *International Journal of Data Mining and Knowledge Management Process* 3, no. 4 (2013): 15–33, <https://doi.org/10.5121/ijdkp.2013.3402>.
- [43] L. Agbota, E. Agyemang, P. Kissi-Appiah, et al., "Enhancing Tumor Classification through Machine Learning Algorithms for Breast Cancer Diagnosis," *Computer Engineering and Intelligent Systems* 15, no. 1 (2024): 71.
- [44] K. Shaukat, S. Luo, V. Varadharajan, I. A. Hameed, and M. Xu, "A Survey on Machine Learning Techniques for Cyber Security in the Last Decade," *IEEE Access* 8 (2020): 222310–222354, <https://doi.org/10.1109/access.2020.3041951>.
- [45] K. Shaukat, S. Luo, V. Varadharajan, et al., "Performance Comparison and Current Challenges of Using Machine Learning Techniques in Cybersecurity," *Energies* 13, no. 10 (2020): 2509, <https://doi.org/10.3390/en13102509>.
- [46] Y. Pristyanto, I. Pratama, and A. F. Nugraha, "Data Level Approach for Imbalanced Class Handling on Educational Data Mining Multiclass Classification," in *2018 International Conference On Information And Communications Technology (ICOIAC)* (IEEE, 2018), 310–314.
- [47] A. Islam, S. B. Belhaouari, A. Ur Rehman, and H. Bensmail, "KNNOR: An Oversampling Technique for Imbalanced Datasets," *Applied Soft Computing* 115 (2022): 108288, <https://doi.org/10.1016/j.asoc.2021.108288>.
- [48] Z. Zheng, Y. Cai, and Ye Li, "Oversampling Method for Imbalanced Classification," *Computing and Informatics* 34, no. 5 (2015): 1017–1037.
- [49] L. I. Santos, M. O. Camargos, M. F. S. V. D'Angelo, et al., "Decision Tree and Artificial Immune Systems for Stroke Prediction in Imbalanced Data," *Expert Systems with Applications* 191 (2022): 116221, <https://doi.org/10.1016/j.eswa.2021.116221>.
- [50] T. Liu, W. Fan, and C. Wu, "A Hybrid Machine Learning Approach to Cerebral Stroke Prediction Based on Imbalanced Medical Dataset," *Artificial Intelligence in Medicine* 101 (2019): 101723, <https://doi.org/10.1016/j.artmed.2019.101723>.
- [51] D. Popescu, M. El-Khatib, and L. Ichim, "Skin Lesion Classification Using Collective Intelligence of Multiple Neural Networks," *Sensors* 22, no. 12 (2022): 4399, <https://doi.org/10.3390/s22124399>.
- [52] L. Hoang, S.-H. Lee, E.-J. Lee, and Ki-R. Kwon, "Multiclass Skin Lesion Classification Using a Novel Lightweight Deep Learning Framework for Smart Healthcare," *Applied Sciences* 12, no. 5 (2022): 2677, <https://doi.org/10.3390/app12052677>.
- [53] M. A. Khan, Yu-D. Zhang, M. Sharif, and T. Akram, "Pixels to Classes: Intelligent Learning Framework for Multiclass Skin Lesion Localization and Classification," *Computers and Electrical Engineering* 90 (2021): 106956, <https://doi.org/10.1016/j.compeleceng.2020.106956>.
- [54] K. Thurnhofer-Hemsi and E. Domínguez, "A Convolutional Neural Network Framework for Accurate Skin Cancer Detection," *Neural Processing Letters* 53, no. 5 (2021): 3073–3093, <https://doi.org/10.1007/s11063-020-10364-y>.
- [55] X. Xing, Y. Hou, H. Li, Y. Yuan, H. Li, and M. Q.-H. Meng, "Categorical Relation-Preserving Contrastive Knowledge Distillation for Medical Image Classification," in *Medical Image Computing and Computer Assisted Intervention-MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part V* 24 (Springer, 2021), 163–173.

- [56] S. S. Chaturvedi, K. Gupta, and P. S. Prasad, "Skin Lesion Analyser: An Efficient Seven-Way Multiclass Skin Cancer Classification Using Mobilenet," in *Advanced Machine Learning Technologies and Applications: Proceedings of AMLTA 2020* (Springer, 2021), 165–176.
- [57] M. Fraiwan and E. Faouri, "On the Automatic Detection and Classification of Skin Cancer Using Deep Transfer Learning," *Sensors* 22, no. 13 (2022): 4963, <https://doi.org/10.3390/s22134963>.
- [58] M. Wang, X. Yao, and Y. Chen, "An Imbalanced-Data Processing Algorithm for the Prediction of Heart Attack in Stroke Patients," *IEEE Access* 9 (2021): 25394–25404, <https://doi.org/10.1109/access.2021.3057693>.
- [59] T. D. Piyadasa and K. Gunawardana, "A Review on Over-sampling Techniques for Solving the Data Imbalance Problem in Classification," *International Journal on Advances in ICT for Emerging Regions (ICTer)* 16, no. 1 (2023): 22–31, <https://doi.org/10.4038/ict.v16i1.7260>.
- [60] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: Synthetic Minority Over-Sampling Technique," *Journal of Artificial Intelligence Research* 16 (2002): 321–357, <https://doi.org/10.1613/jair.953>.
- [61] L. Muflikhah, F. A. Bachtar, D. E. Ratnawati, and R. Darmawan, "Improving Performance for Diabetic Nephropathy Detection Using Adaptive Synthetic Sampling Data in Ensemble Method of Machine Learning Algorithms," *Jurnal Ilmiah Teknik Elektro Komputer dan Informatika* 10, no. 1 (2024): 123–137, <https://doi.org/10.26555/jiteki.v10i1.28107>.
- [62] C. Bunkhumpornpat, K. Sinapiromsaran, and C. Lursinsap, "Safe-Level-Smote: Safelevel-Synthetic Minority Over-Sampling Technique for Handling the Class Imbalanced Problem," in *Advances in Knowledge Discovery and Data Mining: 13th Pacific-Asia Conference, PAKDD 2009 Bangkok, Thailand, April 27-30, 2009 Proceedings 13* (Springer, 2009), 475–482.
- [63] T. Miftahushudur, H. Mertkan Sahin, B. Grieve, and H. Yin, "Enhanced Svm-Smote With Cluster Consistency for Imbalanced Data Classification," in *International Conference on Intelligent Data Engineering and Automated Learning* (Springer, 2023), 431–441.
- [64] G. E. A. P. A. Batista, R. C. Prati, and M. C. Monard, "A Study of the Behavior of Several Methods for Balancing Machine Learning Training Data," *ACM SIGKDD explorations newsletter* 6, no. 1 (2004): 20–29, <https://doi.org/10.1145/1007730.1007735>.
- [65] E. Nyarko, E. F. Agyemang, and D. Arku, "Covid-19 and Online Teaching in Higher Education: A Discrete Choice Experiment," *Model Assisted Statistics and Applications* 18, no. 2 (2023): 115–123, <https://doi.org/10.3233/mas-221403>.
- [66] G. LemaAZtre, F. Nogueira, and C. K. Aridas, "Imbalanced-Learn: A python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning," *Journal of Machine Learning Research* 18, no. 17 (2017): 1–5.
- [67] K. Shaukat, S. Luo, and V. Varadharajan, "A Novel Deep Learning-Based Approach for Malware Detection," *Engineering Applications of Artificial Intelligence* 122 (2023): 106030, <https://doi.org/10.1016/j.engappai.2023.106030>.